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Foundation Models in Medical Image Analysis: Overview and Prospects

Essam A. Rashed
Graduate School of Information Science,
University of Hyogo,
Kobe 650-0047, Japan
rashed@gsis.u-hyogo.ac.jp

Mahmoud Bekhit
Peter Faber Business School,
Australian Catholic University,
North Sydney, Australia
mahmoud.bekhit@acu.edu.au

***Abstract*—Foundation models have emerged as remarkable tools in medical image analysis leading to a new era of smart healthcare and offering sophisticated capabilities to enhance diagnostic accuracy, treatment planning, and patient outcomes. This paper provides a comprehensive overview of the current standing and future prospects of foundation models in the field of medical imaging. We briefly highlight some applications across diverse imaging modalities such as MRI, CT, and ultrasound and discuss limitations and challenges. Future prospective and research directions are discussed to guide future exploration.**

***Index Terms*—Foundation models; deep learning; medical image**

I. Introduction

Medical image diagnosis is a common clinical procedure that aim at using different image modalities in confirmation of disease diagnosis. With the extensive use of medical imaging and the shortage of medical experts, it becomes difficult to keep the traditional data flow, where physician and radiologists are suffering from workload. Artificial intelligence emerge to improve the healthcare ecosystem through robust and fast medical image analysis models. In the past decades, there have been several attempts to develop and implement deep learning models in various medical image analysis applications. However, these models are typically engineered to solve specific problems within particular modalities. Although deep learning models often outperform conventional models and become the state-of-the-art in many clinical applications, they are still far from widespread adoption due to a lack of generality. The diversity in data acquisition and inter-/intra-subject variability limits the use of deep learning models in healthcare facilities (with minor exceptions). A few months ago, the concept of "foundation models" emerged as broad-scope models designed to facilitate universal medical image processing [1]. These models require intensive training with a variety of datasets to extend the training domain to a more general space [2], [3]. A recent study demonstrated that foundation models from the natural domain are useful for medical image classification problems [4]. The main challenge is to build a trustworthy and transparent framework that can gain the trust of medical experts for clinical use [5]. This paper discusses the current progress of foundation models in medical image analysis and provides potential perspectives for future directions.

II. Overview of foundation models

Foundation models (FMs) have contributed to an evolutionary change in medical image analysis by using the previously trained on big-scale datasets; these models are designed to generalize over a range of tasks and techniques, overcoming the constraints of conventional deep learning models that sometimes lack generality (Fig. 1). The ideas of various necessary investigations are combined in this section and connected to using foundation models in medical image analysis.

A. Foundation Models in Medical Imaging

Deep learning is now the leading technology in medical image analysis [6]. However, the current paradigm is that deep learning models are developed to solve narrowly focused problems that are evaluated with limited data. This progress limits the generality and usage of the developed methods on a large scale. A potential solution to this problem is the employment of FMs, where the model is designed to enable flexibility in data and tasks. A major challenge with the medical image analysis is the availability of annotated data required for wide-scale training process. Zhang *et al.* [1] explore the potential, applications, and future directions of major, pre-trained FMs in medical image analysis. The study mentioned that medical FMs could maintain patient data, reduce the large amounts of labeled data requirement, increase model accuracy and robustness across various downstream activities. The authors address their challenges and applications for general imaging, modality-specific, and organ/task-specific FMs. The main challenge with medical image analysis is the variability in comparison with natural images. Therefore, it is difficult to build a unified multi-scale FM to control organ and imaging modalities' topology, shape, size, and texture. Adapting vision FM to domain-specific medical difficulties requires particular algorithms and fine-tuning techniques.

B. Components of Foundation Models

Foundation models in medical image analysis exhibit several components that set them apart from traditional machine learning models. One of the primary characteristics is their scalability, allowing these models to handle vast amounts of diverse data, including various types of medical images such as X-rays, MRIs, and CT scans. This also includes the fuse

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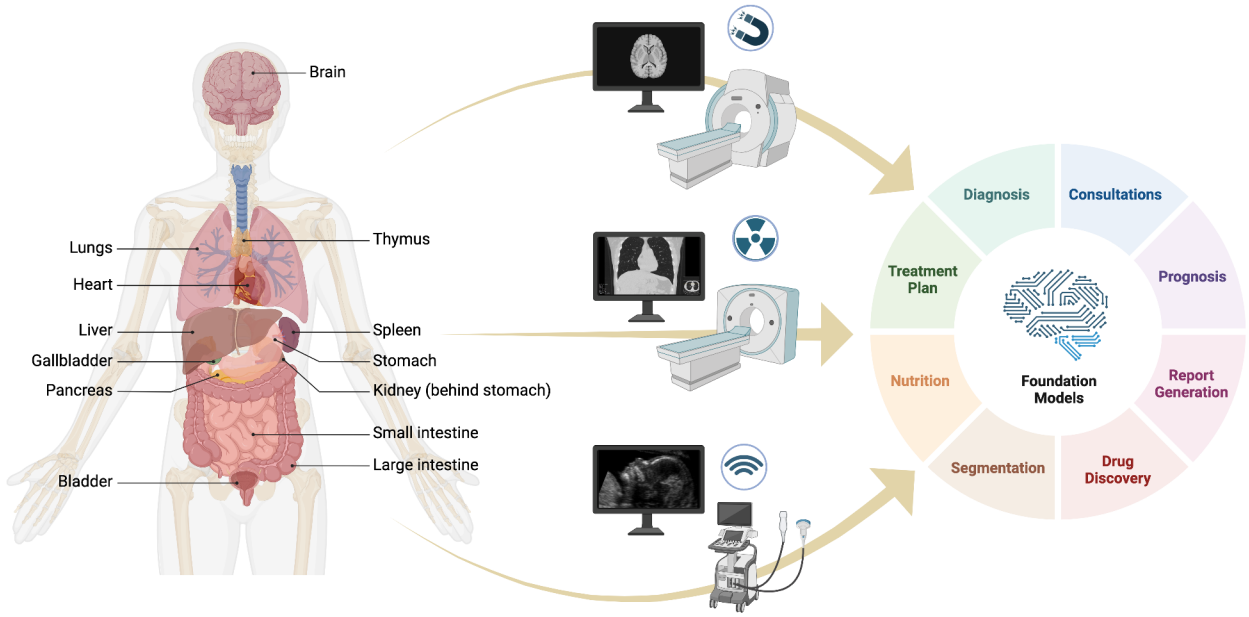

Fig. 1. Foundation models in medical imaging are trained on multiple medical data modalities can be used for comprehensive clinical procedures.

of multi-modality clinical data such as text, sound and data acquired using different types of sensors (e.g., EEG and ECG). This scalability is facilitated by sophisticated network architectures, such as transformers [7] and convolutional neural networks (CNNs) [8], which can capture intricate patterns and features in the data. Additionally, foundation models benefit from transfer learning, enabling them to leverage pre-trained knowledge from large datasets and apply it to specific medical imaging tasks. A common example in this track is to train network using natural images, where annotated data is available, then fine tune with medical images. This technique helps in improving accuracy and reducing the need for extensive domain-specific data.

Another important component of foundation models is their adaptability and flexibility. These models are designed to be fine-tuned for multiple tasks, making them highly versatile in addressing different problems within medical image analysis. For instance, a single foundation model can be adapted for tasks such as image segmentation, disease diagnosis, and anomaly detection, providing a unified solution across various applications. This adaptability is achieved through architecture engineering that allow easy integration of additional layers or components tailored to specific tasks. Furthermore, the use of attention mechanisms within these models enhances their ability to focus on relevant regions of medical images, improving interpretability and diagnostic accuracy. Overall, the combination of scalability, adaptability, and advanced architectural components makes foundation models a powerful tool in the realm of medical image analysis.

C. Advancements in Model Training and Implementation

Advancements in model training and implementation of foundation models in medical image analysis have been dramatically progressing, driven by innovations in neural network architectures, data augmentation techniques, and transfer learning strategies. The authors of [9] conduct a study on the efficacy of Parameter-Efficient Fine-Tuning (PEFT) methods, with a particular emphasis on a new approach known as Embedded Prompt Tuning (EPT). EPT aims to include prompt tokens in expanded channels to adapt foundation models to medical image classification tasks. The investigation determines that prompt tuning facilitates the reduction of intra-class distance and the enhancement of inter-class feature separation. The main limitations of their approach include three data heterogeneity and collection methods, as well as computational and memory costs for universal datasets. Also, the gap in the approach identified is the necessity for additional investigation of EPT in fine-grained perception tasks, such as segmentation. The theoretical investigation of EPT's function in distribution calibration and its application in more intricate medical imaging tasks should be the primary focus of future research. Segment anything model (SAM) appears to performs efficiently in several natural image segmentation tasks [10]. Later on, it is extended to fit with the segmentation requirements in medical images with remarkable performance [11]. A case study of using SAM in several clinical segmentation tasks in reported in [12].

Another study investigate a new deblurring masked image modelling (MIM) method for ultrasonic image analysis [13]. The process is done by adding an image deblurring task to the MIM framework. The proposed approach greatly improves

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the performance of downstream activities, including nodule classification, Hashimoto's thyroiditis classification, and nodule segmentation, by using a hybrid convolution-transformer encoder to acquire fine- and coarse-grained picture representations. The work performs better than supervised and other self-supervised learning methods using significant pretraining on two big ultrasonic datasets: US-GE-10K and US-Esaote-280K. Despite its effectiveness, the method has drawbacks, including limited transferability to ultrasonic imaging of various organs resulting from the thyroid-specific pretraining dataset. The results display that the strategy's efficacy grows with the volume of pretraining data, thus emphasising the importance of big datasets.

Foundation models are also utilized for examining pathology and endoscopic images, particularly for stomach inflammation and cancer [14]. The study underscores the necessity of big data and self-supervision in training these algorithms to enhance prediction accuracy. The results yield robust models for disease segmentation and classification using heterogeneous information. Early detection of stomach cancer facilitated by endoscopic surveillance enhances patient care and treatment outcomes. However, the study has limitations, including the requirement for significant computational resources and the potential for model "hallucinations" leading to misdiagnosis. Another study propose OpenPath, where a large dataset of pathology images paired with natural language descriptions [15]. The proposed model is a fuse of image and text and results indicate that publicly shared medical information is a good resource that can be used for training of FMs. Virchow is a vision transformer with hundred millions parameters trained on millions of digital pathology images [16]. Virchow demonstrate high performance in identification of rare cancer types.

Wang *et al.* proposed a new framework that combines task-specific fine-tuned models with a basic model to improve volumetric medical image segmentation [17]. The framework employs the Mixture of Experts (MoE) technique, which dynamically selects the optimal expert model using a lightweight gating network based on input images and prompts. This approach preserves the segmentation capability of the basic model while enhancing performance on specific tasks. However, despite its advancements, the model has limitations. It may experience 'catastrophic forgetting,' where previously learned knowledge is lost due to fine-tuning on new tasks, requiring significant computational resources. Additionally, the model's reliance on massive datasets highlights challenges related to availability and diversity in medical imaging data. Future research should focus on developing new foundation models and refining the gating network dynamically to improve model adaptability and efficiency.

Jin *et al.* proposed FairMedFM, a complete benchmark, to evaluate the fairness of FMs in medical imaging [18]. The proposed benchmark tests 17 medical imaging datasets and 20 frequently used FMs for downstream applications like segmentation and classification. FairMedFM utilizes a standardized pipeline that includes prompting, parameter-efficient fine-tuning, linear probing, and zero-shot learning. The analysis reveals FM performance biases, utility-fairness trade-offs, and poor bias mitigating techniques. This work underlines the requirement of fairness criteria in medical image analysis foundation models to produce equal results for different patient groups. The difficulty of fixing biases across all sensitive variables and the requirement for significant computational resources define FairMedFM's whole approach's limitations.

Denner *et al.* examine the usage of FMs for content-based medical image retrieval (CBIR) in radiology [19]. Feature extractors without fine-tuning are self-supervised, weakly-supervised, and completely-supervised FMs examined. The study also encourages improved feature extraction, as sick features are more difficult to recover than physical structures. The study has limitations, such as index size-dependent retrieval performance, the difficulty of obtaining minor clinical features due to anatomical similarities and fine-tuning specific medical datasets that might help them.

The privacy hazards connected to post-hoc FM explanations in medical imaging is assessed in [20]. The authors utilize many model architectures, including Vision Transformers (ViTs) pre-trained on big datasets, to image categorization problems. The study underlines how sensitive FMs are to membership inference attacks, which may expose private information on the training data, especially pertinent in medical settings when patient privacy takes the stage. The study found that overfitted models enhance privacy hazards since their interpretations varied considerably across training and non-training data.

Another study proposed a large dataset and benchmark for FM modification in medical image classification [21]. There were a large dataset including chest X-rays for thoracic illness screening, pathology slides for lesion identification, endoscopic images for colorectal lesion identification, digital camera photographs for newborn jaundice evaluation, and retinal images for diabetic retinopathy grading. Among the baseline evaluation techniques are transfer learning like DenseNet, EfficientNet, and Swin Transformer and few-shot learning like Meta-Baseline and Visual Prompt Tuning (VPT). The study indicated that FMs may identify odd illnesses and modify models economically with limited data. Optimizing FMs for visual modalities and characteristics, as well as the demand for significant processing resources, is the study's main limitation. Several surveys have been recently available such as [22]–[24]

III. Challenges and Limitations

Foundation models in medical image analysis face several challenges and limitations that impact their widespread application and effectiveness. One significant challenge is the need for large-scale, diverse datasets for training these models effectively. Due to strict restrictions and guidelines on the usage of medical data, it becomes difficult to acquire enough data that can be used i training purposes. Moreover, medical imaging datasets often vary in quality, size, and annotation

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consistency, which can hinder the generalizability and performance of FMs across different healthcare settings and patient demographics. These challenges underscore the importance of robust data governance frameworks and collaborative efforts to establish comprehensive datasets that support the development and validation of FMs in medical contexts.

Another critical limitation is the interpretability and explainability of FMs. Deep learning models, such as CNNs and transformer-based architectures, often function as black boxes, making it challenging to understand how they arrive at specific diagnostic decisions. This lack of interpretability can impede trust among healthcare providers and patients, limiting the clinical adoption of these advanced technologies. There is always a resistance from medical expert to blindly accept decision made by AI model without having clear explanation on how this decision is reached. Moreover, the computational complexity and resource-intensive nature of training and deploying FMs pose practical challenges, especially for healthcare institutions with limited computational infrastructure and expertise. Addressing these interpretability issues and optimizing model performance in real-world clinical settings are crucial areas of ongoing research and development in the field of medical image analysis.

IV. Future Directions

The future of FMs in medical image analysis holds promise for advancements in many different aspects. For instance, it is expected that there will be shortage in healthcare human resources[1], therefore traditional data processing and handling daily-based tasks in healthcare facilities will be considered challenging. Foundation models can play a significant role in medical data analysis and knowledge extraction with medical experts on the loop. However, several points need to be consider before FMs take a rule in the healthcare ecosystem such as those discussed in previous section. Integrating multimodal data sources, including clinical notes, genetic information, and real-time physiological data, promises to enrich the context and accuracy of diagnostic predictions. Furthermore, the development of federated learning approaches that facilitate collaborative model training across different healthcare institutions while preserving data privacy could democratize access to advanced medical imaging technologies.

V. Conclusion

This paper discuss FMs as a key tool in the progress of medical image analysis. Current research and future prospectives demonstrate that FMs will soon take an important role in enabling a more efficient and feasible healthcare services with remarkable performance complicated data analysis and understanding. However, several limitations and challenges need to be carefully addressed to satisfy ethical constraints regarding the use of medical data.

[1] *https://www.who.int/health-topics/health-workforce#tab=tab_1*

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